

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026

Mathematics — Class 9 (Science Group)

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Total Marks: 43

Time: 2 Hours

Paper: Exam-15 (Version 1)

1. Attempt **ALL** questions. There are **no choices** in this paper.
2. Section A (24 marks): 24 MCQs. Fill the correct bubble. Suggested time: 30 minutes.
3. Show complete working for every question that requires it.
4. Use black or blue ink only. Pencil is allowed only for diagrams.

SECTION A — Multiple Choice Questions (24 × 1 = 24 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	The number $\sqrt{2}$ is:	a rational number	an irrational number	a natural number	a whole number	(A)(B)(C)(D)
2	In the radical $\sqrt[3]{8}$, the number 8 is called the:	radical	radicand	index	exponent	(A)(B)(C)(D)
3	The exponential form of $\sqrt[5]{(x^3)}$ is:	$x^{3/5}$	$x^{5/3}$	x^{15}	x^{-2}	(A)(B)(C)(D)
4	0.333... is an example of a:	terminating decimal	non-terminating recurring decimal	irrational number	whole number	(A)(B)(C)(D)
5	Using the laws of exponents, $a^5 \times a^{-2}$ equals:	a^7	a^3	a^{-10}	$a^{5/2}$	(A)(B)(C)(D)
6	Written in standard form of scientific notation, 45000 is:	4.5×10^3	4.5×10^4	45×10^3	0.45×10^5	(A)(B)(C)(D)
7	The value of $\log_2 8$ is:	2	3	4	8	(A)(B)(C)(D)
8	The integral part of a common logarithm is called the:	mantissa	characteristic	antilogarithm	radicand	(A)(B)(C)(D)
9	A logarithm taken to the base e is called a:	common logarithm	natural logarithm	antilogarithm	characteristic	(A)(B)(C)(D)
10	According to the laws of logarithms, $\log(mn)$ equals:	$\log m \times \log n$	$\log m + \log n$	$\log m - \log n$	$n \log m$	(A)(B)(C)(D)
11	The factors of $a^2 - b^2$ are:	$(a - b)^2$	$(a + b)(a - b)$	$(a + b)^2$	$a^2 + b^2$	(A)(B)(C)(D)
12	The expression $x^2 + 5x + 6$ factorizes to:	$(x + 2)(x + 3)$	$(x + 1)(x + 6)$	$(x - 2)(x - 3)$	$(x + 5)(x + 1)$	(A)(B)(C)(D)
13	The expression $a^3 + b^3$ factorizes to:	$(a + b)(a^2 - ab + b^2)$	$(a + b)(a^2 + ab + b^2)$	$(a - b)(a^2 + ab + b^2)$	$(a + b)^3$	(A)(B)(C)(D)
14	For any two algebraic expressions p and q , the product $\text{HCF} \times \text{LCM}$ equals:	$p + q$	$p \times q$	$p - q$	$p \div q$	(A)(B)(C)(D)
15	Factorizing $ka + kb + kc$ gives:	$k(a + b + c)$	$(k + a)(b + c)$	$k(abc)$	$(ka)(kb)(kc)$	(A)(B)(C)(D)
16	The expression $a^2 + 2ab + b^2$ is the expansion of:	$(a - b)^2$	$(a + b)^2$	$a^2 - b^2$	$(a + b)(a - b)$	(A)(B)(C)(D)
17	The value of $ -7 $ is:	-7	7	0	1/7	(A)(B)(C)(D)
18	The solution of the equation $4x - 12 = 0$ is:	$x = 3$	$x = -3$	$x = 4$	$x = 12$	(A)(B)(C)(D)
19	The property stating that for any real numbers a and b exactly one of $a < b$, $a = b$, $a > b$ is true is called the:	transitive property	additive property	trichotomy property	multiplicative property	(A)(B)(C)(D)
20	Dividing both sides of the inequality $-2x < 6$ by -2 gives:	$x < -3$	$x > -3$	$x < 3$	$x > 3$	(A)(B)(C)(D)
21	If a and b are the coordinates of two points A and B on a coordinate line, the distance $d(A, B)$ is:	$a + b$	$ b - a $	ab	$b \div a$	(A)(B)(C)(D)
22	The distance between the points $(2, 1)$ and $(6, 4)$ is:	5 units	7 units	25 units	10 units	(A)(B)(C)(D)
23	The mid-point of the line segment joining	$(2, 3)$	$(5, 2)$	$(4, 6)$	$(10, 4)$	(A)(B)(C)(D)

	(-3, 5) and (7, 1) is:					
24	Three or more points that lie on the same straight line are said to be:	non-collinear	collinear	parallel	perpendicular	ⒶⒷⒸⒹ

SECTION B — Short Questions ($3 \times 3 = 9$ Marks)

Q.2	Attempt all three parts.	Marks
(i)	(a) Express in exponential form : (i) $4\sqrt{x^5}$ (ii) $3\sqrt{a^2}$. [2] (b) Using the laws of exponents, simplify: $(x^3 \times x^5) \div x^4$. [1]	3
(ii)	(a) Using the laws of logarithms, write as a single logarithm : $\log m + \log n - \log p$. [1] (b) Find the value of $\log_5 125$, showing your working. [2]	3
(iii)	Solve the inequality $7x - 4 \leq 24$ and write the solution set. State which property of inequalities you used at each step. [3]	3

SECTION C — Long Questions ($2 \times 5 = 10$ Marks)

Q.3	Attempt both questions.	Marks
(i)	(a) Factorize completely: $x^2 + 9x + 20$. [1] (b) Factorize completely: $8a^3 + 27b^3$. [2] (c) Find the HCF and the LCM of $8a^2b$ and $12ab^3$, and verify that $\text{HCF} \times \text{LCM}$ equals the product of the two expressions. [2]	5
(ii)	The points A(-1, 2) and B(5, 10) are plotted on the Cartesian plane. (a) Find the distance AB. [2] (b) Find the mid-point M of the line segment AB. [2] (c) Verify that M is equidistant from A and B. [1]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Units Covered: 1, 2, 4, 5 and 7

— End of Question Paper —